Timing a pulsating star with off-the-shelf cameras: DY Pegasi observed by two PANOPTES telescopes

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(30 September 2026)

PANOPTES telescopes are built from ordinary digital cameras. We ask a simple question: can such a telescope measure the pulsation period of a known star as well as the published value? We observed the pulsating star DY Pegasi on 13 nights spread over 19 days in September 2026, with two PANOPTES units in California and three cameras. We wrote a short, fully documented photometry pipeline and first tested it against the survey pipeline AutoWISP on a star where the answer was known. The two agree closely (correlation 0.998 or better). The test also shows a known effect of colour cameras very clearly: the star's image is smaller than one colour cell, so its light moves between the two green pixels as the telescope drifts. Adding the four colour pixels together removes it. From 7 476 images of 30 seconds we measure a period of $P = 0.0729255 \pm 0.0000005$ d, or 105.0127 minutes known to 0.04 seconds. This agrees with the catalogue value to within 1.6 times the uncertainty. The two telescopes, and all three cameras, agree with each other. Each image measures the star's brightness to about 4.5%, and three constant comparison stars show no false signal at the pulsation period. We also detect a second, much weaker pulsation. We then measure 55 times of maximum light and use them to test the best published model of DY Peg's long-term timing, which was fitted to data up to 2019. Our maxima arrive 282 ± 4 s earlier than the model predicts for 2026. Images from the TESS satellite, which we analyse independently, show the maxima already 94 s early in 2022. We refit the model with all 1 068 times of maximum from 1948 to 2026. Two descriptions fit: a steadily shrinking period with a companion orbit longer than 55 years, or a 41-year orbit with a period that has begun to shrink faster. They predict maxima up to 1.5 minutes apart by 2029, so a few more seasons of timings will decide between them.

Keywords: pulsating stars; DY Pegasi; photometry; times of maximum; O−C diagram; low-cost telescopes; PANOPTES

1 Introduction

Some stars change their brightness in a regular rhythm. They swell and shrink, and as they do they grow brighter and fainter. The time one cycle takes is called the *period*. It depends on the star's size, mass and inner structure, so a precise period is a window into the star [1].

Measuring a period does not need a large telescope. It needs many measurements, spread over many nights. This makes it a good task for small robotic telescopes. PANOPTES (Panoptic Astronomical Networked Observatories for a Public Transiting Exoplanets Survey) is a project that builds such telescopes from commercial digital cameras, camera lenses and a small mount [2–4]. The units are built and run by school students, university students and amateur astronomers [4].

Before such a telescope is used for new science, it should pass a test with a known answer. We chose the pulsating star DY Pegasi. Its period has been measured many times over several decades [5–8]. If PANOPTES gives the same period, both the hardware and our analysis are working.

This paper describes the test. Section 2 introduces the star and Section 3 the observations. Section 4 explains how we turn a camera image into a brightness, and how we checked each step. Section 5 explains how we measure the period. Section 6 gives the results, and Section 7 discusses what we learned.

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2 The star: DY Pegasi

DY Pegasi (DY Peg) is a pulsating star of the SX Phoenicis type. These are old stars, a little hotter than the Sun, that pulsate quickly and strongly [1]. DY Peg completes one cycle in about 1 hour and 45 minutes. In that time its brightness in visible light changes from magnitude 10.62 to 9.95 [5].

Astronomers measure brightness in *magnitudes*. A smaller magnitude means a brighter star. A change of 1 magnitude is a factor of 2.5 in brightness, so DY Peg is almost twice as bright at its peak as at its faintest. In this paper we often use *millimagnitudes* (mmag). One mmag is a thousandth of a magnitude, or roughly a change of 0.1% in brightness.

The General Catalogue of Variable Stars lists the period as $P = 0.072926297$ d [5]. Xue and Niu [8] give a pulsation frequency of $f_0 = 13.71249$ cycles per day, or $P = 0.0729262$ d. The period is known to shrink very slowly, by about 6 parts in 10^8 per year [8], and the star orbits a companion once every 42 years [8]. Both effects are far too small to change the period during our 19 days. Over decades, however, they add up to several minutes in the times at which the star is brightest. Xue and Niu [8] fitted 689 such times, from 1947 to 2019, with a model that includes both effects. Because our observations are seven years later, they test how well that model predicts the future (Section 6.7). DY Peg also has two much weaker pulsations, at 17.70 and 18.14 cycles per day [8].

Table 1. The data. "Images kept" are those that passed every quality check and were used to measure the period.

Unit	Camera	Site	Dates (2026)	Images	Images kept
PAN007	95cdbc	Mount Wilson	10–16 Sep	2 661	2 641
PAN015	5205b8	Big Bear	12–29 Sep	2 725	2 689
PAN015	5f1e70	Big Bear	22–29 Sep	2 222	2 191
All			19.3 days	7 608	7 521

3 Telescopes and observations

We used two PANOPTES units in southern California (Table 1). PAN007 is at Mount Wilson Observatory (latitude 34.22° N, 1742 m above sea level). PAN015 is at Big Bear (latitude 34.27° N, 2053 m).

Each unit has two Canon digital cameras in a weatherproof box on an iOptron iEQ30 Pro equatorial mount, which turns to follow the stars. A Raspberry Pi computer runs the unit. Each camera looks through a Rokinon 85 mm lens used fully open, at f/1.4 [9]. The raw files show that both PAN015 cameras are Canon EOS Rebel SL1 bodies (also sold as the 100D), with 5208×3476 pixels, each 4.38 micrometres across. PAN007's camera produces images of the same size and the same scale on the sky, so it is very likely a camera of the same kind; its raw files were no longer available to check. One camera of PAN007 and both cameras of PAN015 recorded DY Peg.

From our star maps (Section 4.3), one pixel spans 10.3 arcseconds of sky, and an image covers $14.8° \times 9.9°$. With 4.38-micrometre pixels, this scale means a focal length of 87.6 mm, close to the lens's nominal 85 mm. Newer PANOPTES units use 24-megapixel cameras with smaller pixels, which give about 9 arcseconds per pixel [10]. Every image was exposed for 30 seconds.

The telescopes observed DY Peg between 10 and 29 September 2026. Some nights were cloudy, and 13 nights gave usable data. In total we downloaded 8 089 images.

Before the analysis we removed 481 images that do not show DY Peg. On the night of 17 to 18 September, PAN007 was pointed 18 to 21 degrees away from the star, although its log said it was on target. This was the first night after the unit's computer restarted. Four images from PAN015 were completely dark, as if the shutter had not opened. We found both problems by working out where each image actually points (Section 4.3).

4 From camera images to brightness

We wrote our own analysis in Python. It is a single notebook that goes from the raw camera files to every figure and number in this paper. It uses Astropy [11], Photutils [12], NumPy [13], SciPy [14] and Matplotlib [15]. It is much smaller than a survey pipeline, because it measures one star instead of thousands.

4.1 Colour pixels, and why we add them together

A digital camera sees colour through a grid of tiny filters. Each block of 2×2 pixels, called a *Bayer cell*, has one red, two green and one blue pixel (Figure 1a). Taking every second pixel therefore gives four separate images, one for each colour.

A star's image on these cameras is very small, about the size of one Bayer cell. This is a known problem for colour cameras. The brightness a star shows in one colour depends strongly on how its image lines up with the pixels of that colour, especially when the image is sharp [10, 16]. Methods for precise photometry with such cameras have been developed for more than a decade [10, 16–18]. The telescope's tracking is never perfect, and the star drifts slowly across the pixels. When it moves by a fraction of a pixel, its light moves from one green pixel to the other (Figure 1b). In a test with five images of the star HD 189733, the brightness in each colour alone scattered by 5 to 11%. The two green channels moved in opposite directions: their correlation was -0.97. The sum of all four colours changed by only 1.0%.

Earlier work has handled this in more elaborate ways. Gee et al. [16] compare each star with other stars whose images fall on the colour grid in the same way. AstroWISP models how light spreads over every part of each pixel [10]. We use a much simpler remedy. We add the light of the four colours before we compute a magnitude: each colour is measured at its own pixel positions, and then the four amounts of light are added. Light that moves from one pixel of a cell to another is then still counted. Measured in one green channel, HD 189733 scattered by 100 to 186 mmag from image to image. With the four colours added, it scattered by only 8 mmag. Penev et al. report better than 1% per colour channel in 2-minute images with their full method [10], so our AutoWISP run of these five images probably did not use those tools to full effect. Adding the colours avoids the problem without any model of the pixels, at the cost of losing colour information. We checked which image is which colour by comparing thousands of stars with their colours from the Gaia satellite: red stars look brighter in the red channel and fainter in the blue one, as they should.

4.2 Measuring a star

To measure a star we add up the light inside a small circle, called an *aperture*, centred on it. The sky itself glows faintly, so we also measure the sky in a ring around the star and subtract it. The brightness in magnitudes is then

$$m = -2.5 \log_{10} F + \text{constant}, \qquad (1)$$

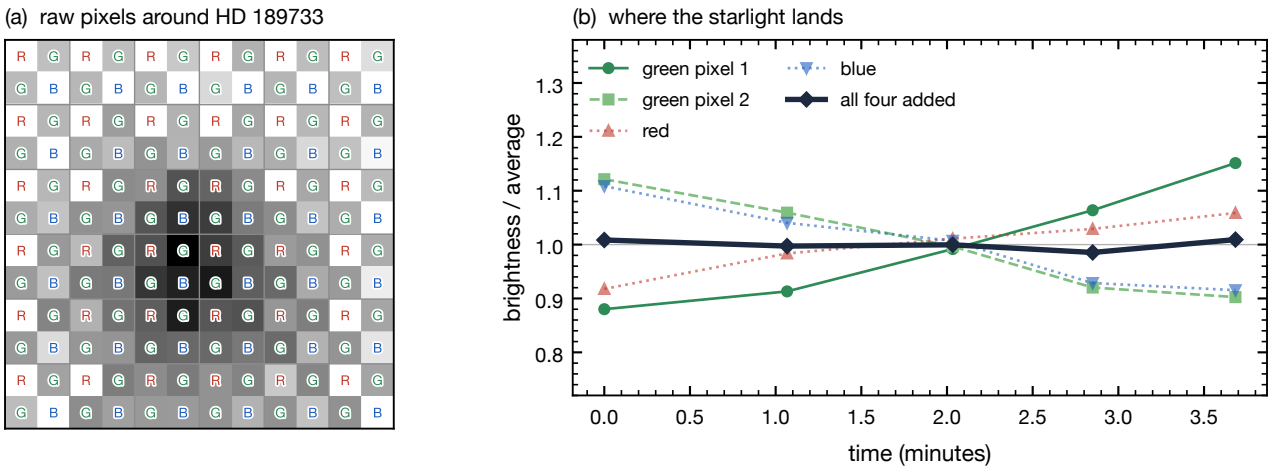


Figure 1. (a) Raw pixels around the bright star HD 189733. Each letter shows the colour of the filter over that pixel; the grid marks the 2×2 Bayer cells. The star's light falls mainly on two or three pixels, about one cell. (b) The star's brightness in each colour over 3.7 minutes, divided by its average. As the telescope drifts by a fraction of a pixel, light moves from one green pixel to the other: one rises by about a quarter while the other falls. The sum of all four colours stays constant to 1%.

where F is the light collected from the star. We tried apertures with radii from 1.5 to 5 pixels of one colour image. We chose the radius using comparison stars (Section 4.4), never using DY Peg itself. The smallest radius, 1.5 pixels (31 arcseconds), gave the least noise. DY Peg is faint enough that a larger circle mostly adds more sky glow.

4.3 Finding the stars

To place the circles, we must know where every star falls in every image. For one image in each sequence of images, the program astrometry.net recognises the pattern of stars and works out where the camera is pointing [19]. We then match about 1 000 stars to their precise positions from the Gaia catalogue [20, 21], taken from the VizieR service [22]. The camera lens bends straight lines on the sky slightly, so we fit a smooth correction for this too. The typical error in a star's position is 0.12 of a Bayer cell, or 2.5 arcseconds. Every other image in the sequence is matched to this one using its brightest 500 stars.

When no solution could be found near DY Peg, we solved the image without any hint of where it should point. This is how we discovered that PAN007 had pointed 18 to 21 degrees away on the night of 17 to 18 September.

4.4 Comparing with other stars

The air above the telescope changes all the time. Thin cloud, haze, and the height of the star above the horizon make every star in an image look fainter or brighter together. We remove this by comparing DY Peg with many other stars in the same image. These are called *comparison stars*.

We chose comparison stars that are within 1.5 magnitudes of DY Peg's brightness, within $4°$ of it on the sky, at least 60 arcseconds from any other catalogued star, and not known to vary [21]. After removing stars that behaved badly on any one camera, 367 remained. For each image, each comparison star is compared with its own usual brightness. The median of these differences tells us how much the air dimmed that image, and we remove it from DY Peg.

These cameras have no *flat field*, the calibration image that corrects for small differences in sensitivity across the sensor. As a result, a star reads slightly differently depending on which pixels it falls on. The telescopes placed DY Peg at eight different spots on the three sensors over the month. We call each of these a *pointing*. We treat each pointing separately: comparison stars are compared with their usual brightness in that pointing, and DY Peg gets one small offset per pointing (Section 5.1).

We also held back three *check stars*. They are as bright as DY Peg and lie within $1.4°$ of it. They are processed exactly like DY Peg but are not used as comparison stars. If the method works, they must come out constant.

Some images were rejected: 3 in which no stars could be matched, 6 with a poor match, 63 in which clouds or twilight removed more than a fifth of the light, 13 that were unusually noisy, and 2 in which DY Peg could not be measured. This left 7 521 images.

4.5 When was each image taken?

Timing matters more than anything else when measuring a period. Each image file carries two times. One is written into the file name by the telescope's computer, which keeps its clock correct over the internet. The other, `DATE-OBS`, comes from the camera's own clock. We found that the camera clocks were set 3 to 7 hours wrong. If `DATE-OBS` were read as Universal Time, DY Peg would often be below the horizon. The file-name time agrees with the height of the star recorded by the telescope to within 0.0002 in airmass for every sequence. We therefore use the file-name time, and add 15 seconds for the middle of the exposure.

Light from DY Peg reaches the Earth up to 8 minutes earlier or later depending on where the Earth is in its orbit.

We correct every time to the moment the light would have reached the centre of mass of the solar system. This is called Barycentric Julian Date ($\mathrm{BJD_{TDB}}$) [23].

4.6 Testing the method where the answer is known

The survey pipeline AutoWISP [10, 24] had already processed five images of the star HD 189733 from PAN007. We ran our method on the same images and compared every step. Our colour images are identical to AutoWISP's, pixel for pixel. The brightness we measure agrees with AutoWISP's to within about 10 mmag. The light curves agree with a correlation of at least 0.998, and differ by only 7 to 18 mmag, while the star itself scattered by 100 to 190 mmag in one colour.

This test found one trap. The two programs count pixel positions differently: one places the centre of a pixel at a whole number, the other half a pixel further on. Without correcting for this, our small apertures missed part of the starlight and read 115 mmag too faint. Differential photometry would have hidden most of this error, because it affects every star in a similar way. Only the direct comparison with AutoWISP revealed it.

5 Measuring the period

5.1 A model of the light curve

A *light curve* is a record of brightness against time. DY Peg's light curve repeats every period P. We describe the position within one cycle by the *phase*

$$\varphi = \frac{t - T_0}{P}, \tag{2}$$

where t is the time and T_0 is a moment of maximum brightness. The whole part of φ counts cycles, and the fraction tells us where we are in the cycle. DY Peg brightens quickly and fades slowly, so its light curve is not a simple sine wave. We model it as a sum of eight sine waves, at the pulsation frequency and its multiples:

$$m(t) = c_p + \sum_{k=1}^{8} \left[a_k \cos(2\pi k\varphi) + b_k \sin(2\pi k\varphi) \right], \tag{3}$$

where c_p is the offset of pointing p (Section 4.4). The numbers a_k, b_k and c_p are found by least squares: they make the model as close as possible to the data, with each point weighted by its uncertainty.

5.2 Three ways to measure the period

Direct fit. We let P be one of the numbers found by the fit.

O−C diagram. For every sequence that covers at least 80% of a cycle, we find the time of maximum brightness by sliding the model in time until it matches the data. We then compare it with the time predicted by the catalogue period. This difference is called O−C, for "observed minus calculated". If the true period differs from the catalogue period by ΔP, the difference grows by ΔP every cycle:

$$\mathrm{O} - \mathrm{C} = \Delta T_0 + E\,\Delta P, \tag{4}$$

where E counts cycles. The slope of a straight line through the points gives ΔP. The scatter of the points about the line tells us honestly how well we know it, including any errors we did not foresee. We use this as our main result.

Expected precision. Montgomery and O'Donoghue [25] give the uncertainty of a frequency measured from N points spread over a time T:

$$\sigma_f = \sqrt{\frac{6}{N}}\,\frac{\sigma}{\pi\,T\,A}, \tag{5}$$

where σ is the scatter of one point and A is the amplitude of the main sine wave. The period uncertainty is then $\sigma_P = \sigma_f\,P^2$. This gives an independent check of our error bar.

6 Results

6.1 The light curve

Figure 2 shows every night of data. The rhythm of DY Peg is clear in every image sequence, and the model follows it closely. On the three nights when both telescopes observed, their measurements lie on top of each other.

Figure 3 shows all 7 476 images folded on the period. Folding means plotting each point against its phase, so that all cycles lie on top of each other. The model uses 7 476 of the 7 521 images; the other 45 were more than four times their expected error from the model and were left out. The star brightens by 0.552 magnitudes from its faintest to its brightest point, a rise of 66%. The scatter of single images about the model is 45 mmag, or about 4.5%. This is 1.3 times more than our error bars predict, so we enlarge all error bars by this factor.

The amplitude we measure is smaller than the 0.67 magnitudes seen in visible light [5]. This is expected. Our cameras add red, green and blue light together, and pulsating stars of this kind change less in red light than in blue.

6.2 Is the method fooling us?

A signal can be created by the analysis itself. For example, a comparison star that varies, or a mistake in the timing, could imprint a false rhythm. The three check stars test this (Figure 4). They were processed exactly like DY Peg. When folded on DY Peg's period they are flat: their averages stay within 20 mmag of zero at every phase, while DY Peg changes by 550 mmag. Their image-to-image scatter is 51 to 61 mmag, similar to that of DY Peg.

6.3 The period

Table 2 and Figure 5 give the results. From all the data we find

$$P = 0.0729255 \pm 0.0000005\ \mathrm{d} = 105.0127 \pm 0.0007\ \mathrm{min}.$$

This is 0.07 seconds shorter than the catalogue value, which is 1.6 times our uncertainty. It is 1.5 times our uncertainty from the value of Xue and Niu [8]. A difference of less than 2 times the uncertainty is normal for measurements that agree, so our period agrees with both.

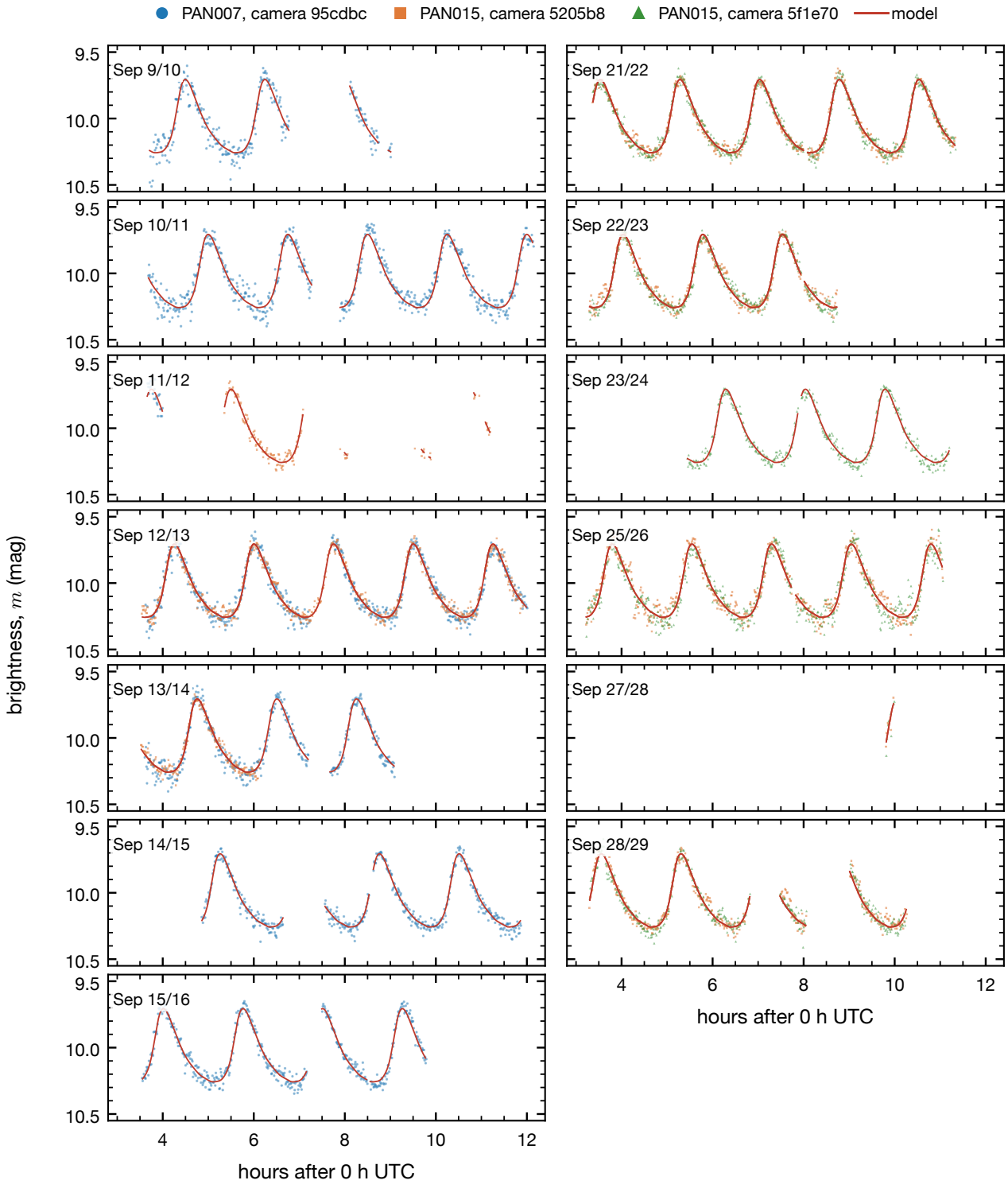


Figure 2. Every night of data, as brightness against time. Colours show the camera. The red line is the model of Equation (3). Magnitudes run downwards, so brighter is higher. On 11 to 14 September both telescopes observed the star at the same time.

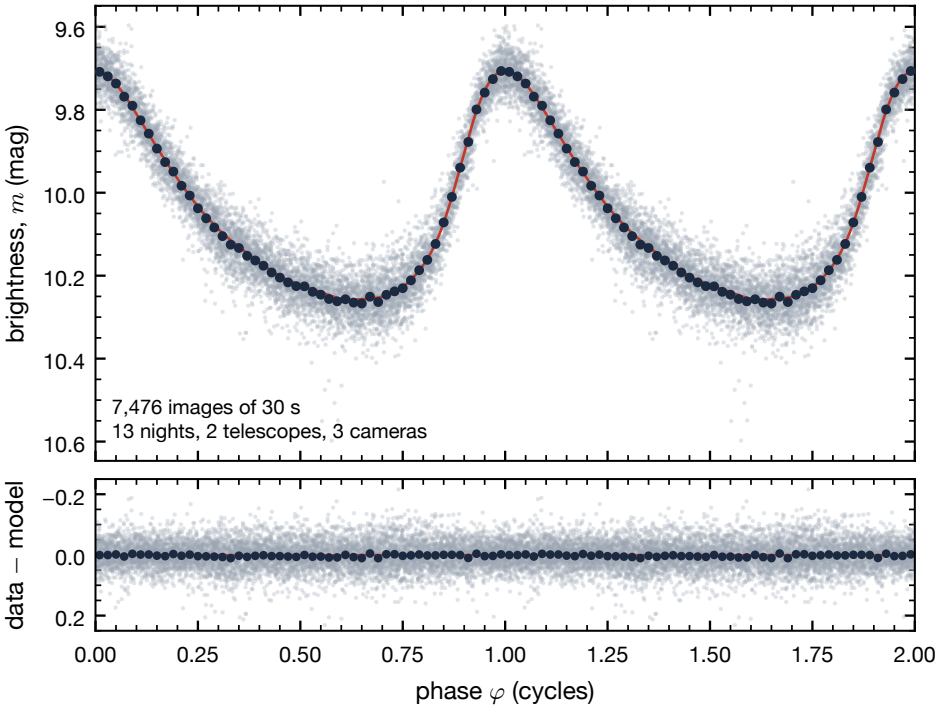


Figure 3. All data folded on the pulsation period, shown twice so that the cycle can be followed. Grey dots are single 30-second images; black points are averages in 50 bins of phase, with their uncertainties. The red line is the model. The lower panel shows the data minus the model.

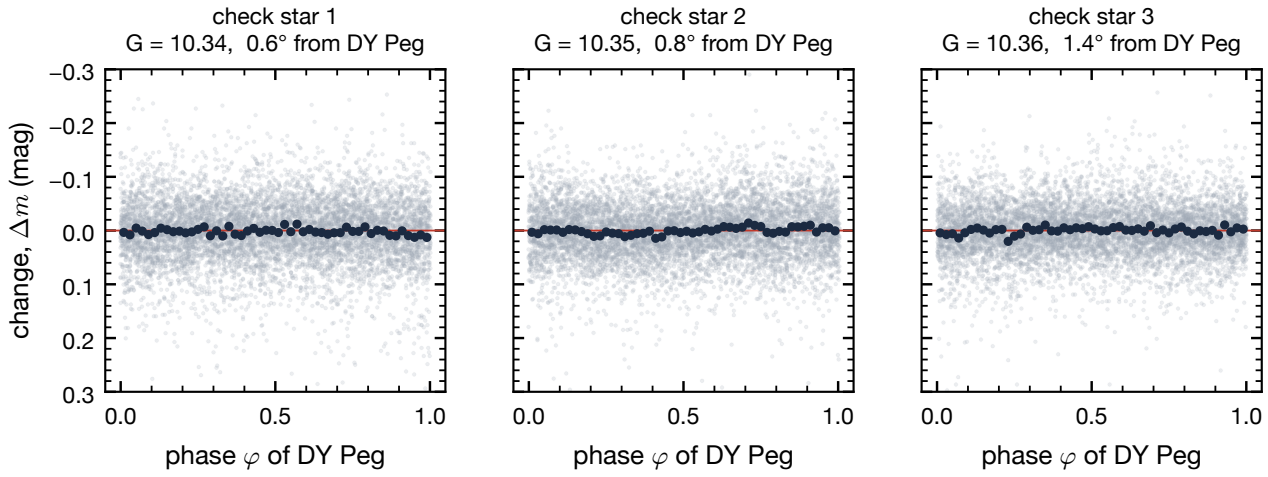


Figure 4. Three constant stars as bright as DY Peg, processed in exactly the same way and folded on DY Peg's period. They show no trace of its rhythm.

Table 2. The pulsation period measured from all data, from each telescope and from each camera. The period and its uncertainty come from the O−C diagram. The last column gives the difference from the catalogue value (0.072926297 d [5]) in units of the uncertainty.

Data	Images	Nights	Days	Period (d)	Difference
All data	7 521	13	19.3	0.0729255 ± 0.0000005	-1.6
PAN007	2 641	7	6.3	0.0729198 ± 0.0000019	-3.5
PAN015	4 880	9	17.2	0.0729256 ± 0.0000007	-1.0
camera 5205b8	2 689	8	17.2	0.0729255 ± 0.0000008	-0.9
camera 5f1e70	2 191	6	7.3	0.0729253 ± 0.0000027	-0.4

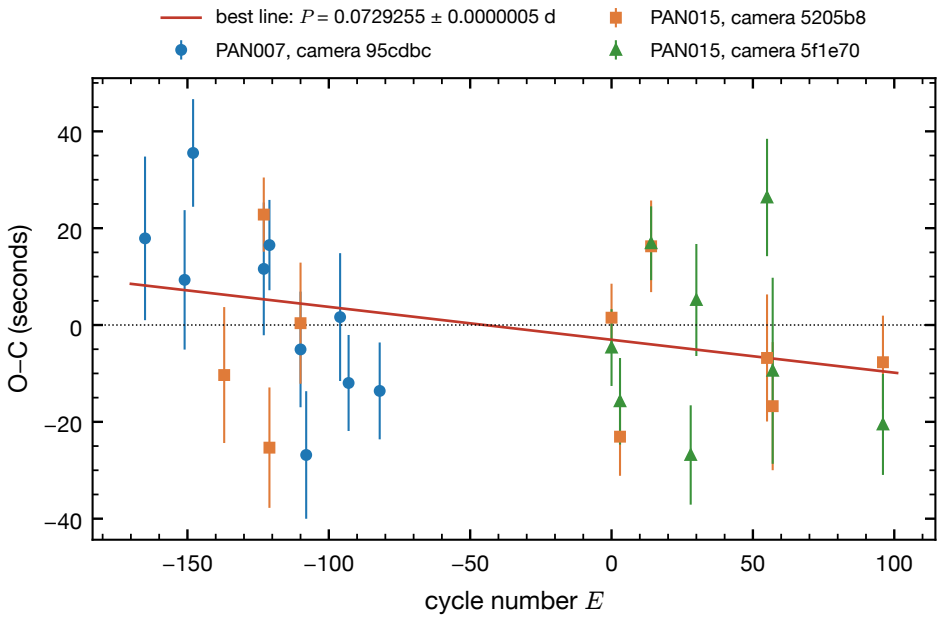


Figure 5. The O−C diagram. Each point is the time of maximum brightness in one sequence of images, minus the time predicted by the catalogue period. The red line is the best straight line; its slope gives the period. A horizontal line would mean the catalogue period is exactly right.

The three ways of measuring the period agree. The direct fit gives 0.0729260 ± 0.0000003 d. Equation (5), with $N = 7\,476$, $\sigma = 45$ mmag, $T = 19.3$ days and $A = 0.248$ mag, predicts an uncertainty of 0.00000045 d. This is almost exactly the 0.0000005 d we find from the O−C diagram. The points in the O−C diagram scatter 1.6 times more than their own error bars suggest; our quoted uncertainty already includes this.

The periodogram in Figure 6 confirms the period without any model. A periodogram tests many trial frequencies and shows how well each one fits the data [26–28]. The highest peak lies at the pulsation frequency. The smaller peaks beside it are *aliases*, echoes one cycle per day away, caused by the daily gaps between nights.

6.4 Three cameras, one star

Each camera sees the same shape of light curve (Figure 7). The amplitudes seen by the two PAN015 cameras agree with the average to within 1%. PAN007 sees an amplitude 2.5% larger. Its camera may differ slightly in colour sensitivity, or its images of DY Peg may be affected by the edge of the field, where it placed the star. Differences of this size do not change the period.

6.5 A second, weaker pulsation

After the main pulsation is removed, the strongest remaining signal is at 16.70 cycles per day (Figure 6, bottom). The chance of a peak this strong arising from noise alone is less than 1 in 10^{20}. It lies exactly one cycle per day away from DY Peg's known second pulsation, at 17.70 cycles per day [8]. Both of our telescopes are in California, so they see the star only during the Californian night. This makes a signal and its one-cycle-per-day echo hard to tell apart. We therefore think we see the known second pulsation, with an amplitude of about 6 mmag. Including it and the third pulsation in the model changes the period by only 0.0000002 d.

6.6 Times of maximum light

Long-term studies of DY Peg are built from *times of maximum light*: the moments at which the star is brightest, collected by many observers over many years [6–8]. We measured these in the standard way. For every maximum that a camera covered well on both the fast rise and the fall, we slid the shape of the light curve in time until it matched the data. This gave 55 times of maximum: 18 from PAN007 and 37 from the two PAN015 cameras. They are listed in Table A1 in Appendix A.

The two PAN015 cameras sit on the same mount, so when both saw the same maximum their two times must agree. On 19 such occasions they disagreed by more than their error bars allowed, because the noise over a few minutes is not independent from image to image. We therefore enlarged all timing errors by a factor of 1.7, so that the pairs agree on average. The median error is then 25 seconds. A straight line through all 55 times gives a period of 0.0729255 ± 0.0000006 d, the same as in Table 2.

6.7 A test of the long-term model

Xue and Niu's model [8] fits their data from 2003 to 2019 well: on average their times lie 14 s from it, with a scatter of 41 s (Figure 8, bottom). Their last maximum is from 2019. Ours are 34 000 cycles later, so we can compare them with the model's prediction without having taken part in the fit.

Our maxima arrive 282 ± 4 s earlier than the model predicts. All three cameras agree: -279, -286 and -280 s.

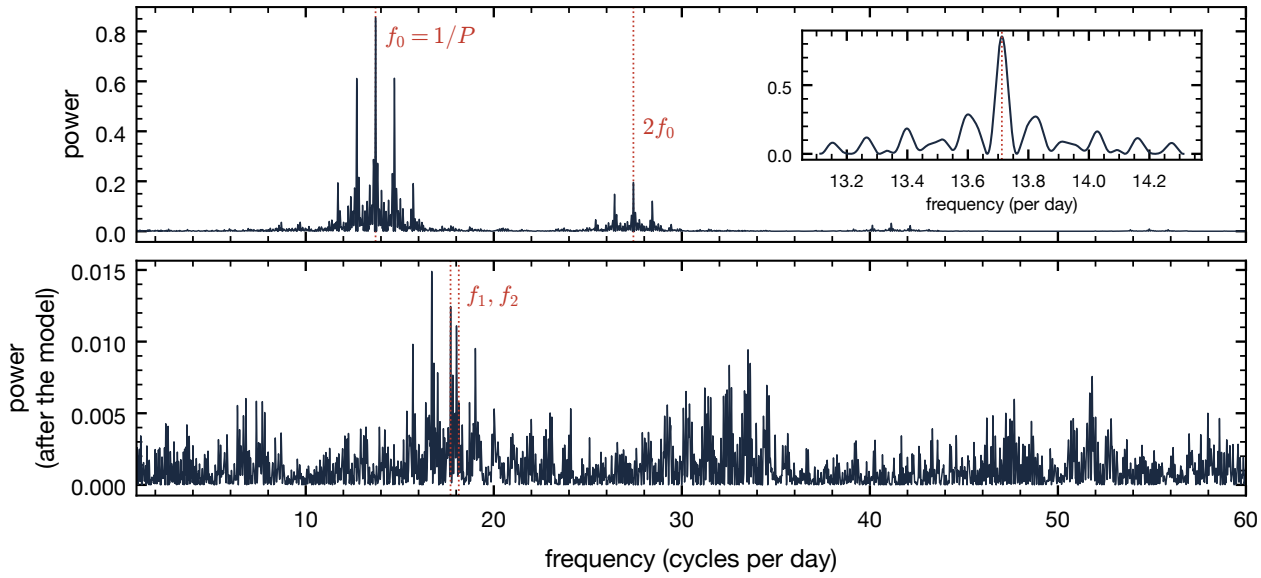


Figure 6. Top: the periodogram of all data. The highest peak is at the pulsation frequency f_0; the inset zooms in on it. Bottom: the periodogram after the main pulsation has been removed. The strongest remaining signal, at 16.70 cycles per day, lies one cycle per day from DY Peg's second pulsation f_1.

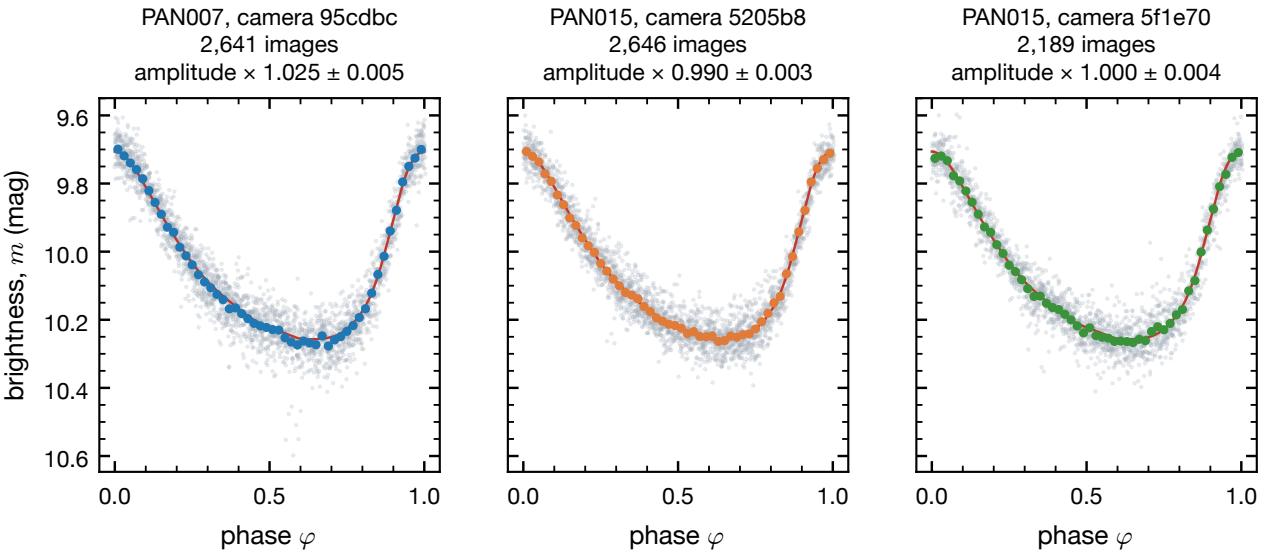


Figure 7. The light curve seen by each camera separately, folded on the period. Coloured points are averages in phase bins; the red line is the model fitted to all data together. The number above each panel is that camera's amplitude relative to the common model.

If each of the model's parameters is varied within its published uncertainty, its prediction for our dates moves by at most about 30 s. Our result is therefore about nine times larger than the combined uncertainty. The older model of Li and Qian [7] does much worse, predicting our maxima 13 minutes late.

A difference of nearly five minutes needs an independent check of our timing. The TESS satellite [29] imaged DY Peg's field in September 2022. There is no ready-made TESS light curve of the star, so we cut out small images around it [30] and measured it ourselves, as for our own images. TESS records a time with every image, independently of our cameras. The correction for the Earth's orbit supplied with the images was computed for another point on the detector; recomputing it for DY Peg changes the times by 18 s, and we use the recomputed times. We measured 338 times of maximum in 26 days, each to about 5 s. On average they arrive 94 ± 1 s earlier than the model predicts. TESS also imaged the field in 2024, but DY Peg fell just outside its detector.

So the maxima follow the model to within about 15 s until 2019, arrive about 1.5 minutes early in 2022, and about 4.7 minutes early in 2026. Two different instruments, with separate clocks, show the same trend.

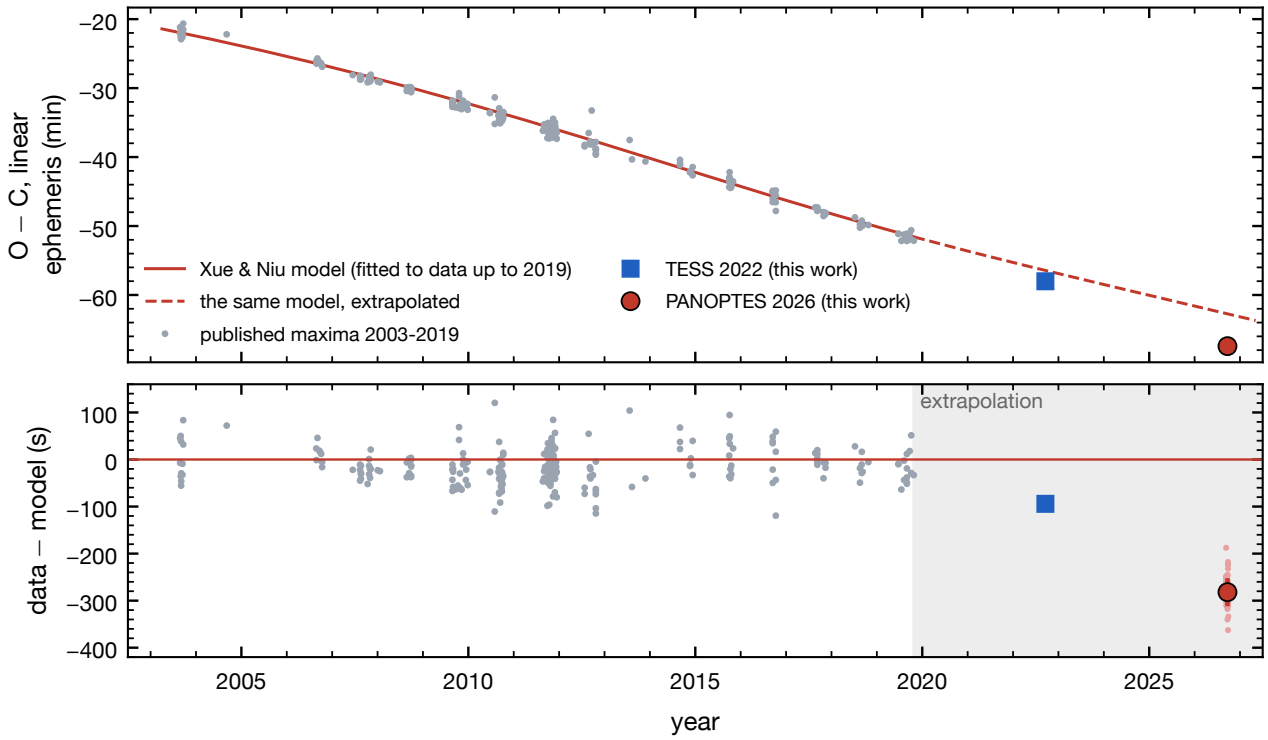


Figure 8. Times of maximum light from 2003 to 2026. Top: each time minus a fixed-period prediction, in minutes. The downward curve is the slowly shrinking period plus the effect of DY Peg's orbit. The red line is the model of Xue and Niu [8], fitted to data up to 2019 (solid) and extended beyond them (dashed). Bottom: each time minus the model, in seconds. Grey: published maxima [8]. Blue: our analysis of TESS images from 2022. Red: our 55 maxima from 2026 (small dots) and their mean.

6.8 A new long-term model

We then refitted the same model to every time of maximum we could find. Li and Qian [7] list 406 from 1948 to 2009, available through VizieR [22]. Xue and Niu [8] list 269 from 2003 to 2019. To these we add our 338 TESS times from 2022 and our 55 PANOPTES times from 2026. That makes 1 068 times over 78 years (Figure 9, top).

As a check, we first fitted only the data up to 2019. This gives almost the same model as Xue and Niu's, with an orbit of 43 instead of 42 years, and it misses the 2022 and 2026 maxima in the same way (by -74 and -245 s).

Three details matter in this fit. First, the published error bars are too small to explain the scatter of the times. We therefore add the same extra uncertainty, 24 s, to every time. Second, the times are not independent: a whole season of maxima is often early or late together, by 17 s on average. The mean times of the 53 seasons scatter about twice as much as their error bars allow, so we divide every difference in χ^2 by 3.8 before comparing models. Third, the fit has many solutions of nearly equal quality. We therefore held the orbital period fixed at values from 33 to 88 years, found the best fit for each, and compared them (Figure 9, bottom).

If the period shrinks at a steady rate, as in Xue and Niu's model, a 42-year orbit no longer fits. The new maxima need an orbit of at least 55 years. Longer orbits, up to at least 88 years, fit equally well, because 78 years of data cover barely one orbit. In this model (fit B) the period shrinks by 6.2 parts in 100 million per year, as before. The 2022 and 2026 maxima are fitted to within 4 and 27 s; the 2026 offset is typical of one season.

The other possibility is that the orbit is about 41 years long, as Xue and Niu found, but that the period has recently begun to shrink faster. This needs one more term in the model, proportional to E^3 (fit C). It fits somewhat better: χ^2 is lower by 14 after the correction, for one extra parameter. In this model the period now shrinks twice as fast as the average over the last 70 years.

We cannot yet choose between these two pictures. They do, however, predict different futures (Figure 9, middle). If fit B is right, maxima in 2027 and later will follow it to within about 10 s. If fit C is right, they will arrive earlier: 22 to 46 s early in September 2027, and 45 to 95 s early in 2029. One season from a PANOPTES unit measures the mean time to about 5 s, and seasons wander by about 17 s. Two or three more seasons should therefore decide.

7 Discussion

7.1 What the early maxima mean

The model of Xue and Niu [8] describes 70 years of timings with two effects: a period that shrinks at a steady rate, and a 42-year orbit around a companion. Our result shows that its extension beyond 2019 fails by several minutes within seven years. The period may be changing at a rate that is not steady, or the orbit may be different from the fitted

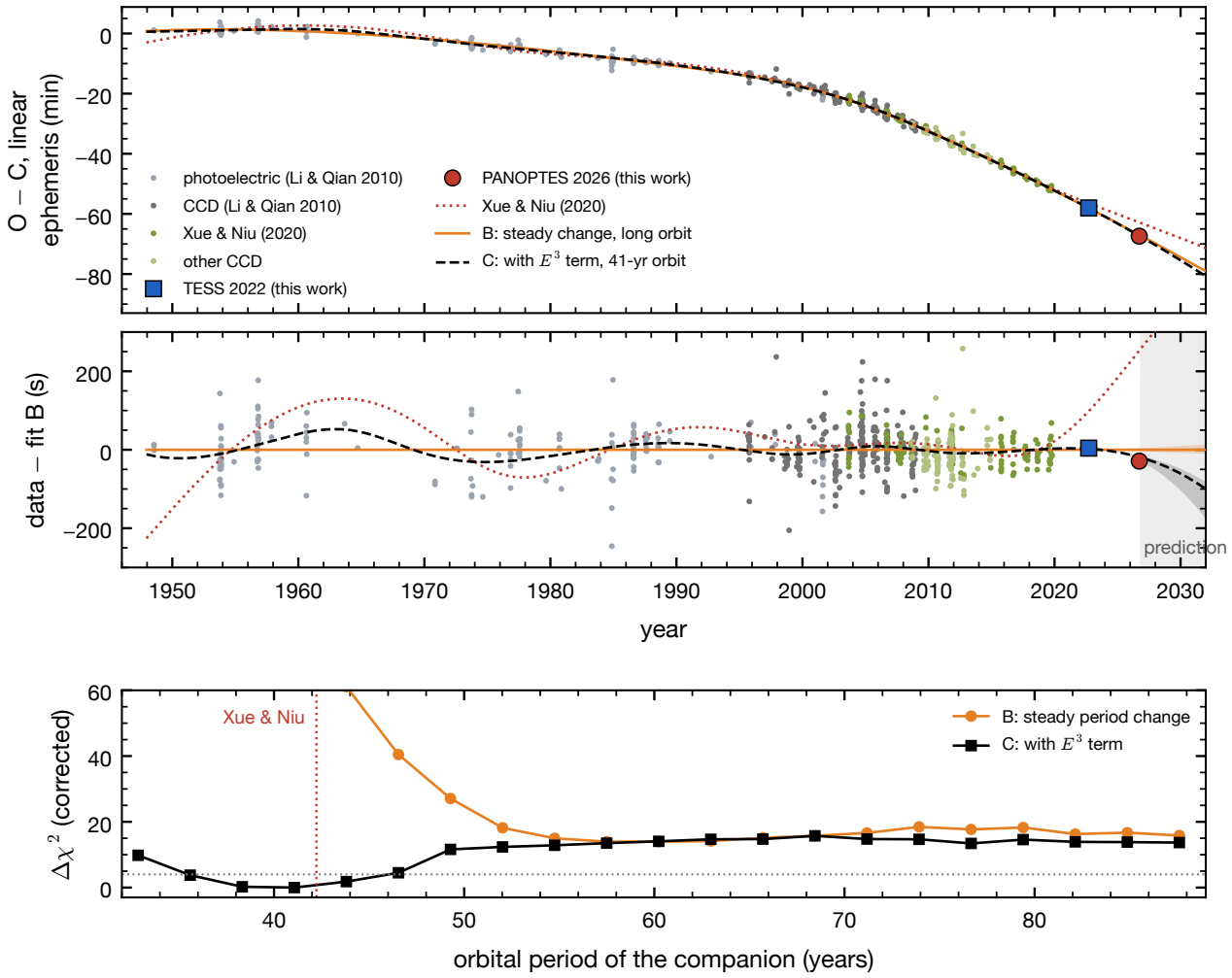


Figure 9. Times of maximum from 1948 to 2026 (1 065 of 1 068; three outliers are left out). Top: each time minus a fixed-period prediction, in minutes, with the model of Xue and Niu [8] (dotted) and our two new fits. Fit B keeps a steadily shrinking period and needs a long orbit. Fit C lets the shrinking speed up and keeps a 41-year orbit. Middle: each time minus fit B, in seconds. The blue square and red circle are the season means of TESS and PANOPTES, with error bars that include the season-to-season scatter. The shaded bands after 2019 show the range of each fit's allowed orbits; after 2027 they are predictions. Bottom: how much worse the fit becomes when the orbital period is held fixed. Values below the dotted line are allowed.

one. Both are plausible, because the orbit has been followed for less than two full cycles, and its fitted shape changed a lot between the studies of Li and Qian [7] and Xue and Niu [8]. Our new fit (Section 6.8) confirms this: both explanations describe all 78 years of data, and neither can yet be excluded.

In an SX Phoenicis star, a steady decrease of the period is usually explained by the slow evolution of the star. Evolution should not change its pace within a few decades, so fit B, with a steady change and a longer orbit, is perhaps the more natural picture. Fit C would need a second cause, such as a second companion or a change in the star's pulsation. We prefer to let the data decide. The predictions in Section 6.8 are specific enough that the next few seasons will do so.

Our times are in the combined colour band of our cameras, and the TESS times in a red band, while most published maxima were measured in visible light. The time of maximum may depend slightly on colour. However, both instruments, in different bands, find maxima that are early, and the difference from the model is three times larger in 2026 than in 2022. Times of maximum measured in visible light in 2025 to 2027 would settle the question.

7.2 Why the first week disagreed

We first analysed only the week of data from PAN007. It gave a period of 0.0729198 ± 0.0000019 d, 3.5 times its uncertainty away from the catalogue value. We searched for a cause. The telescope's clock is kept correct over the internet; even without that, its uncorrected drift of 21 parts per million could explain only a quarter of the difference. The barycentric correction changed by less than one second per day during the run. Including DY Peg's weaker pulsations in the model did not remove the difference either.

With the second telescope and three times the time span, the period agrees with the catalogue. The uncertainty of

a period shrinks roughly as $1/T$ (Equation (5)), so a long time span helps far more than many images on a few nights. PAN007's short result remains unexplained, and it differs from PAN015's by 2.9 times their combined uncertainty. It may be chance, or a small effect we have not found. This is exactly why a known-answer test should use more than one telescope and more than one week.

7.3 Lessons for PANOPTES observers

1. **Add the four colours**, or model the pixels as AstroWISP does. A star's image is smaller than a Bayer cell. One colour alone scattered 10 to 20 times more than the sum in our test.
2. **Check how pixel positions are counted.** A half-pixel mismatch between two programs made small apertures lose over 10% of the light.
3. **Trust the file-name time, not `DATE-OBS`.** The camera clocks were set hours wrong.
4. **Solve every sequence.** After a restart, a telescope reported the right target while pointing 20 degrees away for a whole night. Only matching the stars in the images revealed this.
5. **Observe over many weeks.** Period precision grows with the time span, not with the number of images.
6. **Publish times of maximum.** They are the data that long-term studies need, and even a small telescope measures them to about half a minute.

7.4 Limitations

We have no flat-field images, so a star's brightness depends a little on where it falls on the sensor. We remove this with one offset per pointing; the offsets are small, between -6 and $+26$ mmag. Our magnitudes are in the cameras' own combined colour band, so they cannot be compared directly with visible-light magnitudes. We could use TESS data from only one year, because DY Peg fell off the detector in 2024. Finally, we tested one star. Fainter stars, or stars with smaller changes, will be harder.

8 Conclusions

Two PANOPTES telescopes, built from ordinary digital cameras, measured the pulsation period of DY Pegasi as 0.0729255 ± 0.0000005 d, from 13 nights spread over 19 days. This agrees with the published value. The two telescopes agree with each other, and so do all three cameras. Each 30-second image measures the star to about 4.5%, and constant stars processed in the same way show no false signal. We also see the star's much weaker second pulsation.

The same data give 55 new times of maximum light. They show that the best published model of DY Peg's long-term timing, fitted to data up to 2019, predicts the 2026 maxima 282 ± 4 s too late. Independent TESS images from 2022 show the same trend, at 94 s. This is a genuine new result about the star, not only a test of the telescopes. A new fit to all 1 068 times of maximum since 1948 shows that either the companion's orbit is longer than 55 years, or the period has begun to shrink faster. The two pictures predict different times of maximum from 2027 onward.

The analysis needed three steps that are easy to miss: adding the four colour pixels, matching pixel conventions between programs, and taking the time from the file name. With these, PANOPTES passes a known-answer test. The next steps are to observe DY Peg for longer, including from our units in Bhutan (about 12 hours of longitude from California, which removes the daily echoes in the periodogram), to time its maxima every season until the two long-term models can be told apart, and to turn to stars whose periods are not yet known.

Data and code

All analysis is in one notebook, `dypeg_analysis.ipynb`. It runs from the raw camera files to every figure and number in this paper, and it also contains the test against AutoWISP and the analysis of the TESS images. Our times of maximum and those from TESS are also given as tables (`times_of_maximum.csv`, `tess_times_of_maximum.csv`), and all 1 068 times used in the long-term fit, with their sources and residuals, in `all_times_of_maximum.csv`.

Acknowledgements

We thank the PANOPTES project and the hosts of the PAN007 and PAN015 units at Mount Wilson and Big Bear. This work has made use of data from the European Space Agency (ESA) mission *Gaia*, processed by the Gaia Data Processing and Analysis Consortium (DPAC). This research has made use of the VizieR catalogue access tool, CDS, Strasbourg, France. This paper includes data collected by the TESS mission, which are publicly available from the Mikulski Archive for Space Telescopes (MAST).

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Appendix A: Times of maximum light

Table A1. Our times of maximum light of DY Peg, in $\mathrm{BJD_{TDB}}$ (subtract 68.8 s for HJD in UTC). σ is the uncertainty after the enlargement described in Section 6.6. Camera A is PAN007 (Mount Wilson); B and C are the two PAN015 cameras (Big Bear).

$\mathrm{BJD_{TDB}} - 2460000$	σ (s)	Cam.	$\mathrm{BJD_{TDB}} - 2460000$	σ (s)	Cam.	$\mathrm{BJD_{TDB}} - 2460000$	σ (s)	Cam.
1293.68728	47	A	1298.71921	23	A	1306.81424	21	C
1293.76063	38	A	1298.93773	25	A	1306.81430	21	B
1294.70813	37	A	1299.66719	27	A	1307.76191	26	C
1294.78156	33	A	1299.74005	25	A	1307.90793	22	C
1294.85523	30	A	1299.88579	24	A	1309.65781	41	C
1294.92704	33	A	1305.64654	31	C	1309.65838	45	B
1296.67728	24	B	1305.64712	24	B	1309.73014	35	B
1296.67734	40	A	1305.71986	24	C	1309.73181	39	C
1296.75046	33	A	1305.72010	21	B	1309.80458	34	B
1296.75050	24	B	1305.79312	18	B	1309.80467	35	C
1296.82350	20	B	1305.79349	23	C	1309.87658	35	B
1296.89585	24	B	1305.86573	24	B	1309.87709	41	C
1296.89631	23	A	1305.86613	21	C	1309.95006	46	B
1296.96908	27	A	1305.93850	21	C	1309.95053	78	C
1296.96911	36	B	1305.93864	17	B	1312.64752	40	C
1297.69792	33	A	1306.66786	24	C	1312.72081	23	B
1297.69793	23	B	1306.66824	39	B	1312.72110	19	C
1297.77150	28	A	1306.74091	25	B			
1297.84387	23	A	1306.74151	21	C			